

JUSTIN FOSTER

Saint Louis, MO | 816-341-1680 | fosterj096@gmail.com | github.com/fosterj096 | linkedin.com/in/justin-foster-904805199

SUMMARY

Software engineer and primary technical owner of the enterprise .NET platform running Rheem Manufacturing's parts operations. Sole engineer on its on-prem-to-Azure migration, architect of its authentication and API layer, and technical authority for two client-side engineers. U.S. Air National Guard veteran.

TECHNICAL SKILLS

Languages / Frameworks: C#, VB.NET/VB6, Python, SQL/T-SQL, ASP.NET Core Razor Pages, Entity Framework Core, JavaScript/Chart.js, .NET Framework 4.0 → .NET 8/9

Cloud & DevOps: Azure (VMs, Blob Storage, Azure SQL, Bastion, IIS), AWS S3, Azure DevOps (Pipelines, Repos, Boards, Dashboards), YAML multi-stage CI/CD, self-hosted build agents

Architecture & Auth: Dependency injection, custom middleware, HTTP client architecture, distributed caching, background job processing, structured logging, ASP.NET Identity, Microsoft SSO / Entra ID, OAuth2, JWT issuance & validation

EXPERIENCE

Software Engineer 2 — Simplicity Software (contracted to Rheem Manufacturing)

Feb 2023 – Present

LEADERSHIP & DELIVERY

- Primary technical contact for the platform across Rheem. Scope work with, and defend architecture and delivery decisions to, product managers, warehouse managers, and senior IT leadership.
- **Technical authority for two client-side Rheem engineers.** Scope and assign work in Azure DevOps Boards, gate all changes through pull request review and branch policy, and run hands-on walkthroughs covering architecture, deployment, and tooling.
- Own the **Azure DevOps practice end to end:** boards, branch policy, the team wiki, and the leadership dashboards on work item status, sprint progress, build health, and defect trends, now standing references in planning meetings.

CLOUD MIGRATION & INFRASTRUCTURE

- **Sole engineer on the platform's full on-prem-to-Azure migration.** Azure SQL does not support cross-database queries, so consolidated **5 production databases** into one, rebuilding the schema and rewriting every dependent C# call site and stored procedure before cutover.
- Built a table-driven console application syncing on-prem SQL to **Azure SQL** nightly, keeping Azure environments current through a phased migration within cost and memory constraints.
- Partnered with Rheem's infrastructure and network teams to stand up **DEV and PROD in Azure** (four application servers plus dedicated build agents), mirroring on-prem IIS configuration across all of them with WebDeploy and provisioning runtimes and SSL bindings.
- Migrated asset storage from **AWS S3 to Azure Blob Storage:** wrote the migration tooling, a service serving blob image URLs from the application database, and a scheduled job reconciling against S3 for missing files.
- Enabled **horizontal scaling** by moving shared state off individual servers, centralizing encryption keys in Azure Blob and adding SQL-backed distributed caching for sessions and tokens, so user sessions survive restarts, deployments, and failover.

CI/CD & AUTOMATION

- Consolidated **100+ standalone console applications** into a single parameter-driven .NET project, then authored the DEV/PROD templates and **100+ scheduled pipelines** running them on cron, replacing ad-hoc manual execution with auditable, logged automation.
- Authored the **YAML multi-stage CI/CD pipelines** deploying the platform to dev and prod and provisioned the self-hosted agents they run on.

AUTHENTICATION, APIS & MODERNIZATION

- Rebuilt authentication to support **three login paths:** legacy OAuth2 credentials, ASP.NET Identity credentials on the current distributor platform, and **Microsoft SSO / Entra ID** for Rheem employees. Added pass-through auth accepting a bearer token from either upstream API, plus custom cookie handling for legacy compatibility.
- After an upstream credential loss forced users onto the newer identity provider, built the **shared-token bridge** keeping the legacy API reachable, using per-user locking and request coalescing to prevent duplicate refreshes, with failure handling separating invalid credentials from transient network faults. **Now retiring it:** replacing every remaining legacy API call so the bridge is deleted rather than maintained.
- Own the **API service layer** integrating with Rheem's distributor-facing platform, with centralized token attachment and automatic re-authentication on expiry across login, authorization, and product/serial endpoints.
- Modernizing a codebase originating in **2008** (.NET Framework / VB6). Converted numerous legacy VB applications to C#, several rebuilt as Razor Pages that retired standalone tools; currently leading the API modernization targeting **.NET 8/9**.
- Own the three systems the warehouse runs on daily: receiving & fulfillment with partial-fulfillment and shortage escalation (**~1M parts** since launch), shipping label generation with multi-team approval and PDF output, and a Chart.js reporting dashboard.
- Cut page load times by **30–40%** on key pages by eliminating redundant database round trips and adding route- and query-level caching on the highest-traffic pages.

Project Engineer — TransPerfect

Sep 2022 – Feb 2023

- Wrote Python automation scripts testing voice-assistant response quality on mobile devices against simulated commands.

Aerospace Ground Equipment Technician — Air National Guard

Sep 2014 – Mar 2021

- Maintained and repaired ground support equipment (diesel generators, mobile lighting units) while attending school full-time.

EDUCATION

University of Missouri–Kansas City — B.S. Computer Science, 2022